# Deterministic Optimization

## Linear Optimization Modeling

## Handling Nonlinearity

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## Robust Regression using LP

# Modeling using Linear Programs

**Learning Objectives for this Lesson**

- Reformulate convex PWL by LP
- Introduce Robust Regression model
- Reformulate RR as LP

# Minimizing Convex PWL as LP

- Minimization of a convex PWL function:
  - $\min_{x \in X} f(x) = \max\{a_1^\top x + b_1, \dots, a_m^\top x + b_m\}$

  can be reformulated by introducing a new variable and putting objective into constraint as follows:

- An equivalent reformulation:

- $\min_{x,z} z$

  s.t. $f(x) \leq z$

  $x \in X$

- $\min_{x \in X, z} z$

  s.t. $\max\{a_1^\top x + b_1, \dots, a_m^\top x + b_m\} \leq z$

- $\min_{x \in X, z} z$

  s.t. $a_i^\top x + b_i \leq z \ \ \forall i = 1, \dots, m.$

This is an LP!

# Absolute Value Function

- Absolute value function $|x| \leq z$ can be reformulated as
  - $\max\{x, -x\} \leq z$
  - $x \leq z,\ -x \leq z$
  - Or equivalently, $-z \leq x \leq z$

- If we have $\left|a^\top x - b\right| \leq z$, then we can reformulate as
  - $-z \leq a^\top x - b \leq z$

# Nonlinear Optimization Model

- $\min\ \ 4(|x_1 - x_2| + |y_1 - y_2|) + 1.1(|x_1 - 300| + |y_1 - 1200|) + 0.7(|x_1| + |y_1 - 600|) + \ 0.65(|x_2| + |y_2 - 600|) + +0.4(|x_2 - 600| + |y_2|)$

- $\min 4(z_1 + z_2) + 1.1(z_3 + z_4) + 0.7(z_5 + z_6) + 0.65(z_7 + z_8) + 0.4(z_9 + z_{10})$

  s.t. $-z_1 \le x_1 - x_2 \le z_1$

  $-z_2 \le y_1 - y_2 \le z_2$

  $-z_3 \le x_1 - 300 \le z_3$

  ….

# Linear Classification

- Given a set of training data $(x_i, y_i)$ for $i = 1, \dots, N$, where $x_i$ is an $n$-dimensional feature vector and $y_i$ is a label of value either $0$ or $1$.
- Think about each $x_i$ represents a vector of lab test data of a patient $i$ and $y_i$ labels if this person has a certain disease.
- We want to build a linear classifier, i.e. a linear function
    - $f(x) = b_0 + \sum_{j=1}^{n} b_j x_j$

  so that, for a given feature vector $x$, if $f(x) \geq 0.5$, then $x$ is classified as $y = 1$, otherwise classified as $y = 0$.

# Robust Regression

- The classifier should be optimal in minimizing total prediction error on training data.
- The prediction error is a measure of **distance** between classifier's output and the true label. So there are different ways to measure distance.
- If we use $l_1$-metric, we will have the following **robust regression** model

$$\min_{b_0,\ldots,b_n} \sum_{i=1}^{N} \left| b_0 + \sum_{j}^{n} b_j x_{ij} - y_i \right|$$

# LP Model for Robust Regression

- Robust regression model:

$$\min_{b_0,\ldots,b_n} \sum_{i=1}^{N} \left| b_0 + \sum_{j}^{n} b_j x_{ij} - y_i \right|$$

- LP reformulation:

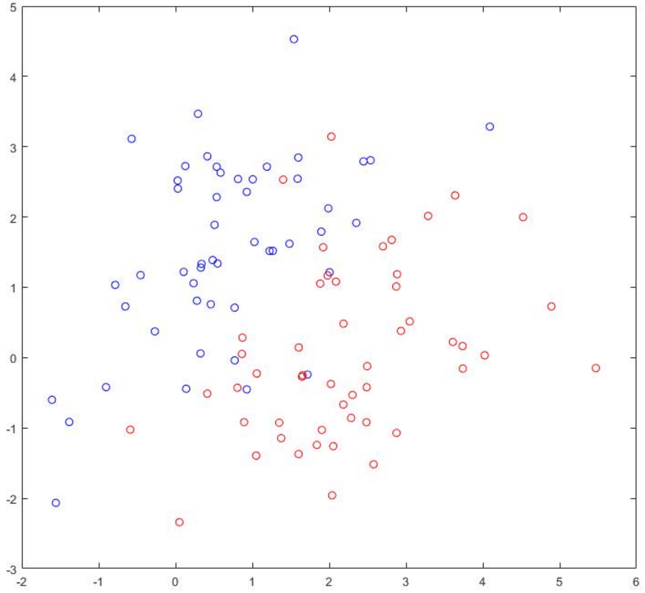
$$\min_{b_0,\ldots,b_n} \sum_{i=1}^{N} z_i$$

$$\text{s.t. } b_0 + \textstyle\sum_j^n b_j x_{ij} - y_i \le z_i \quad \forall i = 1, \ldots, N$$

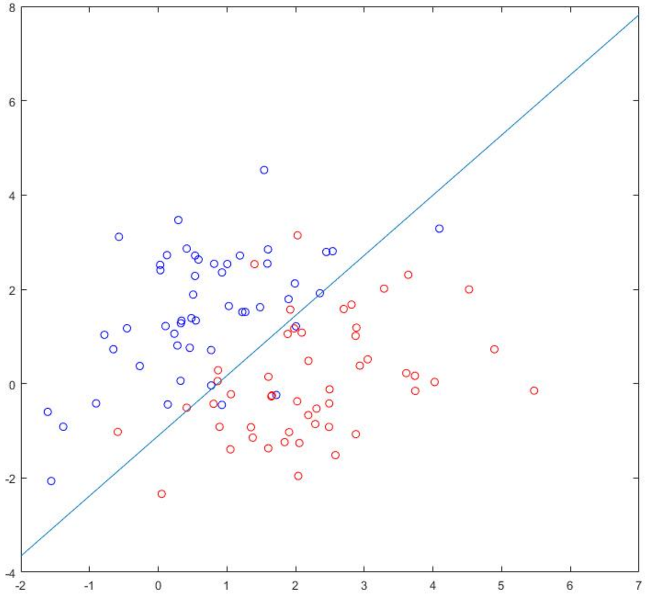
$$b_0 + \textstyle\sum_j^n b_j x_{ij} - y_i \ge -z_i \ \forall i = 1, \ldots, N$$

# Example

Training data




Robust Regression Classifier



# Summary

- We introduced a general way to reformulate any convex PWL minimization as an LP.

- We applied this technique to robust regression, a useful data analysis tool.